ELEC60004
EE9-AML2-07

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE/EIE PART III/IV: MEng, BEng and ACGI

MACHINE REASONING

Monday, 2 December 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J. Pitt
Second Marker(s) : C. Qin

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Information for Students

KE Calculus

α-Formula Rules

$$\frac{p \wedge q}{\begin{array}{c} p \\ q \end{array}} \qquad \frac{\neg(p \vee q)}{\begin{array}{c} \neg p \\ \neg q \end{array}} \qquad \frac{\neg(p \rightarrow q)}{\begin{array}{c} p \\ \neg q \end{array}}$$

β-Formula Rules

$$\frac{p \rightarrow q}{\begin{array}{l} p \\ \quad q \end{array}} \qquad \frac{p \rightarrow q}{\begin{array}{l} \quad \neg q \\ \neg p \end{array}} \qquad \frac{p \vee q}{\begin{array}{l} \neg p \\ \quad q \end{array}} \qquad \frac{p \vee q}{\begin{array}{l} \quad \neg q \\ p \end{array}} \qquad \frac{\neg(p \wedge q)}{\begin{array}{l} p \\ \quad \neg q \end{array}} \qquad \frac{\neg(p \wedge q)}{\begin{array}{l} \quad q \\ \neg p \end{array}}$$

Double Negation Elimination

$$\frac{\neg(\neg p)}{p}$$

η-Formula Rules

$$\frac{p \leftrightarrow q}{\begin{array}{l} p \\ \quad q \end{array}} \qquad \frac{p \leftrightarrow q}{\begin{array}{l} \neg p \\ \quad \neg q \end{array}} \qquad \frac{p \leftrightarrow q}{\begin{array}{l} \quad q \\ p \end{array}} \qquad \frac{p \leftrightarrow q}{\begin{array}{l} \quad \neg q \\ \neg p \end{array}} \qquad \frac{\neg(p \leftrightarrow q)}{\begin{array}{l} p \\ \quad \neg q \end{array}} \qquad \frac{\neg(p \leftrightarrow q)}{\begin{array}{l} \quad q \\ \neg p \end{array}} \qquad \frac{\neg(p \leftrightarrow q)}{\begin{array}{l} \neg p \\ \quad q \end{array}} \qquad \frac{\neg(p \leftrightarrow q)}{\begin{array}{l} \quad \neg q \\ p \end{array}}$$

Branching Rule (PB)

$$\begin{array}{ccc} & \diagup\diagdown & \\ p & & \neg p \end{array}$$

Closure Rule

$$\frac{\begin{array}{c} p \\ \neg p \end{array}}{\times}$$

γ and δ Formula Rules (Predicate Logic)

γ rule	$\dfrac{\forall x.A}{A_{\{c/x\}}}$	$\dfrac{\neg\exists x.A}{\neg A_{\{c/x\}}}$	for any constant c
δ rule	$\dfrac{\exists x.A}{A_{\{c/x\}}}$	$\dfrac{\neg\forall x.A}{\neg A_{\{c/x\}}}$	for a constant c new to the branch

Modal Formula Rules (Modal Logics K and S5)

K

$\Box$-rule	$\dfrac{\sigma : \Box p}{\sigma n : p}$	$\dfrac{\sigma : \neg\Diamond p}{\sigma n : \neg p}$	for any available σn
$\Diamond$-rule	$\dfrac{\sigma : \Diamond p}{\sigma n : p}$	$\dfrac{\sigma : \neg\Box p}{\sigma n : \neg p}$	for σn unrestricted

S5

$\Box$-rule	$\dfrac{i : \Box p}{j : p}$	$\dfrac{i : \neg\Diamond p}{j : \neg p}$	for any available integer j
$\Diamond$-rule	$\dfrac{i : \Diamond p}{j : p}$	$\dfrac{i : \neg\Box p}{j : \neg p}$	for j unrestricted

1. This question is about search space formalisation and the General Graph Search Engine (GGSE), and 'uninformed' search algorithms.

a) Explain, with the help of an example, how a state space can be represented (explicitly and implicitly) as a graph, and how an implicit graph representation can be specified in Prolog, so that it can be used for planning and problem-solving with the GGSE.

[8]

b) (The Jealous (Heteronormative) Husbands Problem (JHB).) Three husbands with their three wives want to cross a river, using a rowing boat that can carry at most two people. The rowing boat cannot cross the river on its own. Moreover, no woman (wife) can be in the presence of (on the same bank as) another man who is not her husband (i.e., there must either be no men on the same bank, or her husband must be on the same bank).

Given that we want to know precisely who rows the boat across, specify an appropriate state representation, and justify this state representation with respect to alternative representations for river-crossing problems of this kind.

[4]

c) Given this state representation, formulate the state space so that it could be used with the GGSE:

(*i*) Specify the initial state;
(*ii*) Specify the final state; and
(*iii*) Specify the state-change rules.

Given the symmetry, only the state-change rules for moving one wife, two wives, and one wife plus one husband from the left bank to the right bank need to be specified.

[6]

d) Explain how your state-change rules use `findall/3` and backtracking to generate all possible next states.

[2]

2. This question is about search space formalisation and the General Graph Search Engine (GGSE), and 'informed' search algorithms.

a) Explain the difference between the Best First and A^* search algorithms.

[2]

b) Explain how the A^* search algorithm expands the paths in a graph defined as $G' = \langle S, op \rangle$.

Prove that the solution path returned by the A^* search algorithm is optimal.

[4]

c) In the context of A^* search, explain the conditions that need to be satisfied for $h(G)$, if G is a goal state, and for h to be monotonic and admissible.

Explain why these conditions might make one heuristic perform "better" than another.

Comment on this explanation with respect to optimality and complexity of the A^* search algorithm.

[6]

d) In the 8-Puzzle, aka the *ah*-Puzzle, the tile locations are labelled $[0..8]$, with locations $0,1,2$ labelling the top row, $3,4,5$ the middle row, etc. The blank can then 'move' (i.e., swap places with) any tile that is in a location connected to the blank's location: for example, if the blank is in location 0, then it can swap with whatever is in location 1 or location 3, The problem state can then represented by a list of 9 integers, the first integer represents the location of the blank tile, the second integer the location of the *a* tile, etc.

A good heuristic is to count Manhattan distance; and an even better heuristic is to count the Manhattan distance plus the number of adjacent reversals (i.e. swapping two adjacent tiles that are in each other's locations), times two.

Explain why this is an admissible heuristic.

Specify a predicate `ith/3` that for an integer index $i, i \geq 0$ and a list, 'returns' the element at the *i*th position in the list, i.e.:

```
| ?- ith( 0, [a,b,c], E ).

E = a;

no

| ?- ith( 3, [a,b,c], E ).

no
```

Hence, or otherwise, specify a predicate that counts the number of adjacent reversals (times two) in a list representing a state of the *ah*-Puzzle.

(You do not need to specify predicates for counting Manhattan Distance.)

[8]

3. This question is about knowledge representation and reasoning with logic and Prolog.

a) In the context of Machine Reasoning, define *resolution*, and state the resolution rule.

[4]

b) In the context of Machine Reasoning, define *unification*. Explain why unification is relevant to resolution, and give a unification algorithm.

[4]

c) Consider the following English statements:

Everyone who has ball-control skills and team-working skills is a footballer.
Everyone who is fit and fast is an athlete.
Everyone who is an athlete has physique.
Everyone who is a footballer has technique.
Everyone who has physique and technique is a goal-machine.
Bryan is fit and fast, and has ball-control skills and team-working skills.

Express these statements in first-order predicate logic, and convert them to a set of Horn clauses.

[4]

d) Prove, using resolution and proof by negation, that *Bryan is a goal-machine*. The proof should show all the unifiers.

[4]

e) Explain what it means for an algorithm for deciding the satisfiability of a logical formula to be *sound* and *complete*.

[4]

4. This question is about reasoning in Propositional Logic, Predicate Logic, and Propositional Modal Logic, using the **KE** calculus.

Important: any answer to any part that doesn't use line numbers and rule references to indicate how new formulas have been derived will not get full marks.

a) Prove, using the **KE** calculus for Propositional Logic, that the following statements are true:

(*i*) $\{a, (a \wedge b) \rightarrow c, \neg c, \neg b \rightarrow (d \wedge e)\} \vdash \neg b \wedge (d \wedge e)$
(*ii*) $\{e \rightarrow g, (e \rightarrow g) \rightarrow (g \rightarrow f), e\} \vdash f$
(*iii*) $\{\neg(r \rightarrow m), (r \wedge e) \rightarrow \neg t, e\} \vdash \neg(t \vee m)$
(*iv*) $\{a \rightarrow b, b \rightarrow a, e\} \vdash (a \leftrightarrow b) \wedge e$

[8]

b) Translate the following syllogism into Predicate Logic. Use **KE** to prove that the conclusion follows from the premises: *All owners of stately homes are country gentlemen. Boris is the owner of a stately home. Therefore Boris is a country gentleman.*

Translate the following syllogism into Predicate Logic. Use **KE** to build a model which makes the premises true and the conclusion false: *Some knowledge is power. Some power is measured in Watts. Therefore some knowledge is measured in Watts.*

[6]

c) Using the **KE** calculus for Propositional Modal Logic, prove that the characteristic formulas for reflexivity, symmetry and transitivity relations are theorems of logic **S5**, but not of logic **K**:

(*i*) $\Box P \rightarrow P$
(*ii*) $P \rightarrow \Box\Diamond P$
(*iii*) $\Box P \rightarrow \Box\Box P$

[6]

5. This question is about formulating a state space for 2-player games using the General Graph Search Engine (GGSE), and the Simplified Event Calculus (EC).

One variant of the Game of Nim is played with a pile of matches, and two players *red* and *blue* who take it in turns to remove 1, 2 or 3 matches from the pile. The winner is the player to take the last match(es).

a) Making some reasonable assumptions about which player takes the first turn and the initial number of matches in the pile, formulate the state space so that it could be used with the GGSE:

(*i*) Define a state representation;
(*ii*) Specify the initial state;
(*iii*) Specify the final state; and
(*iv*) Specify the state-change rules.

[8]

b) Making some reasonable assumptions about which player takes the first turn and the initial number of matches in the pile, specify the Game of Nim as `initially/1` facts about identified multi-valued fluents.

Then define `initiates/3` axioms of the Simplified EC for reasoning about the effects of any action or actions played in the game on these fluents.

These axioms should ensure that a move is a valid move (i.e., the same player cannot take consecutive turns, cannot take more than three matches, cannot take a negative number of matches, etc.).

[8]

c) Explain, using the Game of Nim as an example, how knowledge is represented and reasoned with in the Simplified Event Calculus compared to the state-space formulation as a graph, and briefly discuss the implications for planning to solve problems of this kind.

[4]